

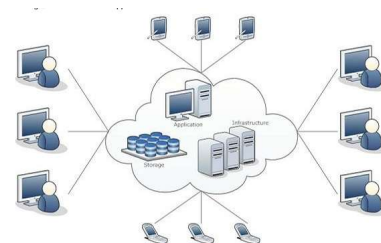
CLOUD COMPUTING

- Cloud Computing refers to manipulating, configuring, and accessing the applications online. It offers online data storage, infrastructure and application.
- The term Cloud refers to a Network or Internet. Cloud is something, which is present at remote location. Cloud can provide services over network, i.e., on public networks or on private networks, WAN, LAN or VPN.
- Applications like e-mail, web conferencing, customer relationship management (CRM) run in cloud.

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CLOUD COMPUTING

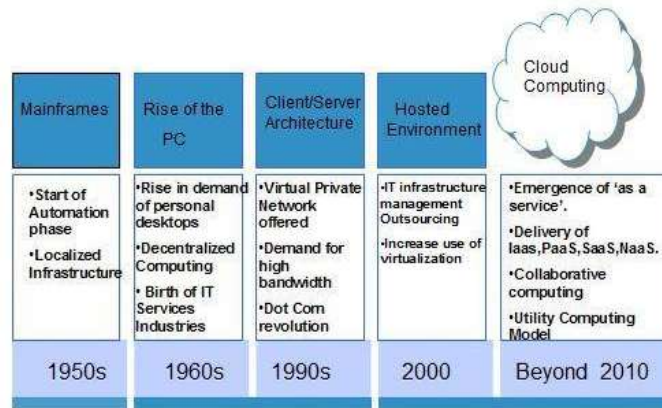
- The delivery of computing, storage and software as a service
- “Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction...”. **The NIST Definition of Cloud Computing.**
- an evolving paradigm....
- “Internet-based computing“
- refers to manipulating, configuring, and accessing the applications online. It offers online data storage, infrastructure and application.
- the Cloud Computing is making business applications mobile and collaborative.



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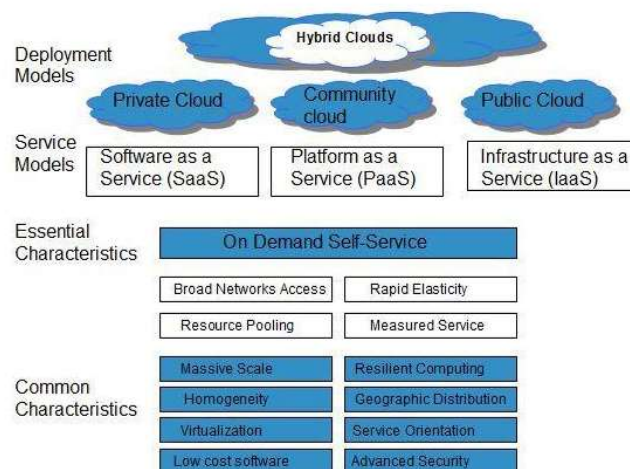
PERSPECTIVE HISTORIC

Accessible via static clients. Since then, cloud computing has been evolved from static clients to dynamic ones from software to services.



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CLOUD COMPUTING



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CLOUD COMPUTING CHARACTERISTICS

The cloud model is composed of five essential characteristics:

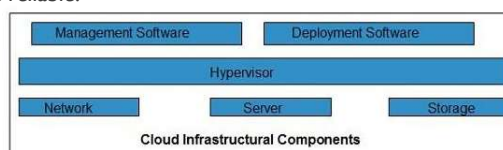
- **On-demand self-service** - A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.
- **Broad network access** - Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).
- **Resource pooling** - The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand. There is a sense of location independence in that the customer generally has no control or knowledge over the exact location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or datacenter). Examples of resources include storage, processing, memory, and network bandwidth.
- **Rapid elasticity** - Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be appropriated in any quantity at any time.
- **Measured service** - Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.

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CLOUD COMPUTING INFRASTRUCTURE

Cloud infrastructure consists of servers, storage, network, management software, and deployment software and platform virtualization.

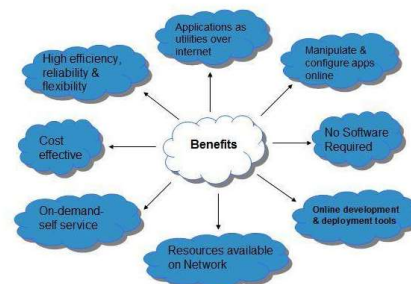
- **HYPERVERSOR** - is a firmware or low-level program that acts as a Virtual Machine Manager. It allows to share the single physical instance of cloud resources between several tenants.
- **MANAGEMENT SOFTWARE** - helps to maintain and configure the infrastructure.
- **DEPLOYMENT SOFTWARE** - helps to deploy and integrate the application on the cloud.
- **NETWORK** - is the key component of cloud infrastructure. It allows to connect cloud services over the Internet. It is also possible to deliver network as a utility over the Internet, i.e., the consumer can customize the network route and protocol.
- **SERVER** - helps to compute the resource sharing and offer other services such as resource allocation and deallocation, monitoring resources, security, etc.
- **STORAGE** - Cloud uses distributed file system for storage purpose. If one of the storage resource fails, then it can be extracted from another one which makes cloud computing more reliable.



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BENEFITS

- can access applications as utilities, over the Internet.
- Manipulate and configure the application online at any time.
- It does not require to install a specific piece of software to access or manipulate cloud application.
- offers online development and deployment tools, programming runtime environment through Platform as a Service model.
- Cloud resources are available over the network in a manner that provides platform independent access to any type of clients.
- offers on-demand self-service. The resources can be used without interaction with cloud service provider.
- is highly cost effective because it operates at higher efficiencies with greater utilization. It just requires an Internet connection.
- offers load balancing that makes it more reliable.

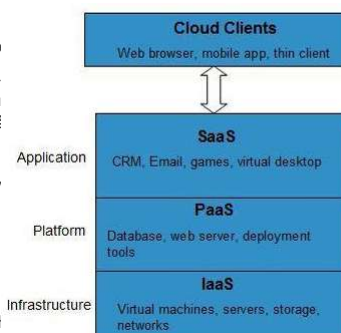


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CLOUD COMPUTING: SERVICE MODELS

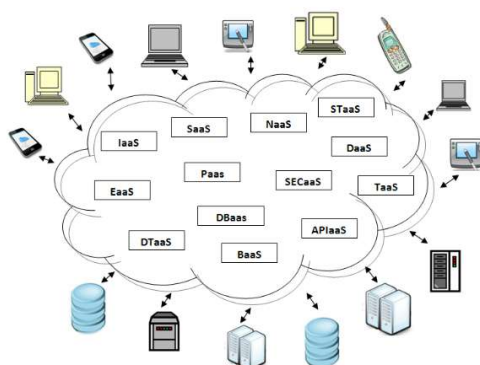
Service Models are the reference models on which the Cloud Computing is based.

- **Software as a Service (SaaS)** - The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure. The applications are accessible from various client devices through either a thin client interface, such as a web browser (e.g., web-based email), or a program interface. The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user specific application configuration settings.
- **Platform as a Service (PaaS)** - The capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages, libraries, services, and tools supported by the provider.³ The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly configuration settings for the application-hosting environment.
- **Infrastructure as a Service (IaaS)** - The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources with the consumer is able to deploy and run arbitrary software, which can include operating systems and applications. The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, and deployed applications; and possibly limited control of select networking components (e.g., host firewalls).
- ...



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A CONCEPTUAL VIEW OF CLOUD COMPUTING



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CLOUD COMPUTING: DEPLOYMENT MODELS

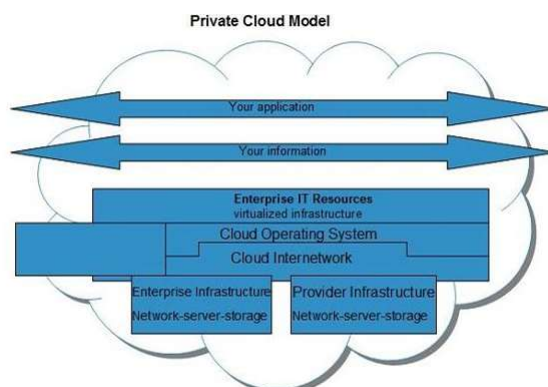
Deployment models define the type of access to the cloud, i.e., how the cloud is located? Cloud can have any of the four types of access:

- **PUBLIC CLOUD** - infrastructure is provisioned for open use by the public. It may be owned, managed, and operated by a business, academic, or government organization, or some combination of them. It exists on the premises of the cloud provider. Public cloud may be less secure because of its openness, e.g., e-mail.
- **PRIVATE CLOUD** - is provisioned for exclusive use by a single organization comprising multiple consumers (e.g., business units). allows systems and services to be accessible within an organization. It offers increased security because of its private nature.
- **COMMUNITY CLOUD** - allows systems and services to be accessible by group of organizations that have shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination of them, and it may exist on or off premises.
- **HYBRID CLOUD** - is mixture of public and private cloud. However, the critical activities are performed using private cloud while the non-critical activities are performed using public cloud.

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PRIVATE CLOUD MODEL

Allows systems and services to be accessible within an organization. It is operated only within a single organization. However, It may be managed internally or by third-party.



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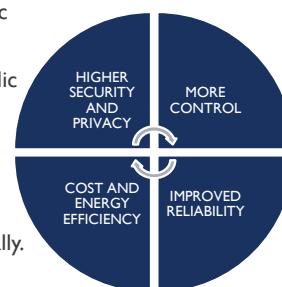
PRIVATE CLOUD MODEL

Benefits:

- **HIGHER SECURITY AND PRIVACY** - Private cloud operations are not available to general public and resources are shared from distinct pool of resources, therefore, ensures high security and privacy.
- **MORE CONTROL** - Private clouds have more control on its resources and hardware than public cloud because it is accessed only within an organization.
- **COST AND ENERGY EFFICIENCY** - Private cloud resources are not as cost effective as public clouds but they offer more efficiency than public cloud. benefits:

Disadvantages:

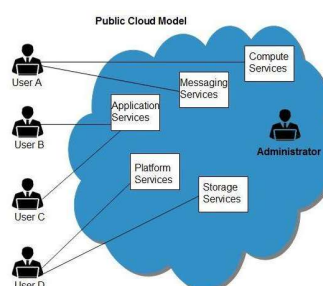
- **RESTRICTED AREA** - Private cloud is only accessible locally and is very difficult to deploy globally.
- **INFLEXIBLE PRICING** - In order to fulfill demand, purchasing new hardware is very costly.
- **LIMITED SCALABILITY** - Private cloud can be scaled only within capacity of internal hosted resources.



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PUBLIC CLOUD MODEL

The Public Cloud allows systems and services to be easily accessible to general public, offers cloud services via Internet, e.g., Google, Amazon, Microsoft offers cloud services via Internet.



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PUBLIC CLOUD MODEL

Benefits

- **Cost effective** - since public cloud share same resources with large number of consumer; it has low cost.
- **Reliability** - Since public cloud employs large number of resources from different locations, if any of the resource fail, public cloud can employ another one.
- **Flexibility** - is also very easy to integrate public cloud with private cloud, hence gives consumers a flexible approach.
- **Location independence** - since, public cloud services are delivered through Internet, therefore ensures location independence.
- **Utility Style Costing** - public cloud is also based on pay-per-use model and resources are accessible whenever consumer needs it.
- **High Scalability** - Cloud resources are made available on demand from a pool of resources, i.e., they can be scaled up or down according the requirement.

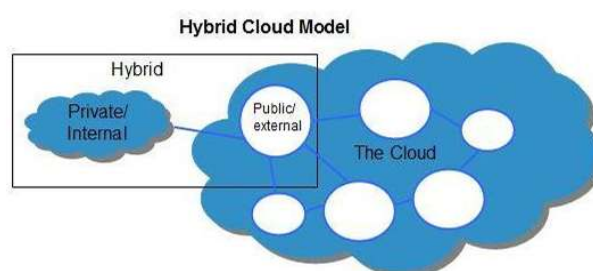
Disadvantages

- **Low Security** - data is hosted off-site and resources are shared publicly, therefore does not ensure higher level of security.
- **Less Customizable** - less customizable than private cloud.

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HYBRID CLOUD MODEL

Hybrid Cloud is a mixture of public and private cloud. Non-critical activities are performed using public cloud while the critical activities are performed using private cloud.



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HYBRID CLOUD MODEL

Benefits:

- **SCALABILITY** - It offers both features of public cloud scalability and private cloud scalability.
- **FLEXIBILITY** - It offers both secure resources and scalable public resources.
- **COST EFFICIENCIES** - Public cloud are more cost effective than private, therefore hybrid cloud can have this saving.
- **SECURITY** - Private cloud in hybrid cloud ensures higher degree of security.

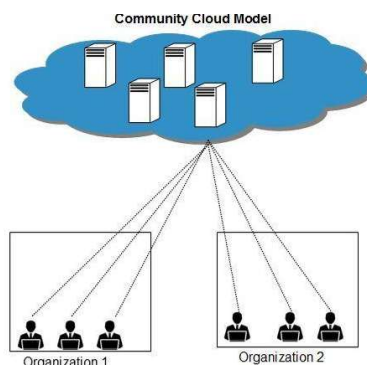
Disadvantages:

- **NETWORKING ISSUES** - Networking becomes complex due to presence of private and public cloud.
- **SECURITY COMPLIANCE** - It is necessary to ensure that cloud services are compliant with organization's security policies.
- **INFRASTRUCTURAL DEPENDENCY** - The hybrid cloud model is dependent on internal IT infrastructure, therefore it is necessary to ensure redundancy across data centers.

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COMMUNITY CLOUD MODEL

- Community Cloud allows system and services to be accessible by group of organizations. It shares the infrastructure between several organizations from a specific community. It may be managed internally or by the third-party.



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HYBRID CLOUD MODEL

Benefits:

- COST EFFECTIVE** - Community cloud offers same advantage as that of private cloud at low cost. Sharing Between Organizations. Community cloud provides an infrastructure to share cloud resources and capabilities among several organizations.
- SECURITY** - Community cloud is comparatively more secure than the public cloud.
- SECURITY** - Private cloud in hybrid cloud ensures higher degree of security.

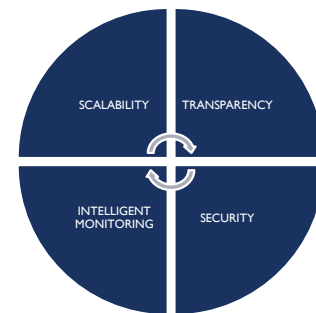
Disadvantages:

- Since all data is housed at one location, one must be careful in storing data in community cloud because it might be accessible by others.
- It is also challenging to allocate responsibilities of governance, security and cost.

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INFRASTRUCTURAL CONSTRAINTS

- **TRANSPARENCY** - Since virtualization is the key to share resources in cloud environment. But it is not possible to satisfy the demand with single resource or server. Therefore, there must be transparency in resources, load balancing and application, so that we can scale them on demand.
- **SCALABILITY** - Scaling up an application delivery solution is not that easy as scaling up an application because it involves configuration overhead or even re-architecting the network. So, application delivery solution is need to be scalable which will require the virtual infrastructure such that resource can be provisioned and de-provisioned easily.
- **INTELLIGENT MONITORING** - To achieve transparency and scalability, application solution delivery will need to be capable of intelligent monitoring.
- **SECURITY** - The mega data center in the cloud should be securely architected. The control node, a entry point in mega data center also needs to be secure.



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CLOUD PLATFORMS

- Cloud platforms are services that provide computing resources and software over the internet.
- A Cloud platform hosts the server hardware and operating system in a web-based datacenter. The platform enables the coexistence of hardware and software, and it offers remote connectivity to compute services at scale.
- Different cloud platforms with its own features, services, and certifications. Most popular:
 - Amazon Web Services (AWS);
 - Microsoft Azure
 - Google Cloud Platform (GCP),
 - IBM Cloud
 - Salesforce
 - Oracle Cloud,
 - SAP BTP e o Alibaba Cloud
 - ...



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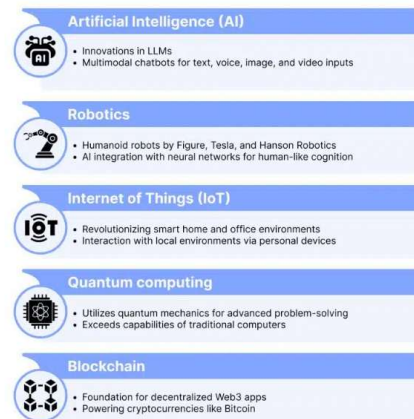
ACTIVITY

- Cutting-edge techniques
- Anything as a Service (XaaS)
- MLaaS: Machine Learning as a Service
- Edge Computing
- Differences Edge Computing/Cloud Computing
- Grid Computing
- Utility Computing
- Virtualization
- Multi-tenant infrastructure.

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CUTTING EDGE TECHNOLOGIES

- **Technological devices, Techniques or achievements that employ the most current and high level IT developments.**
- Cutting-edge technology is important because it introduces new solutions that can enhance human society.
 - is a technology with the latest features and capabilities.
 - AI, robotics, and blockchain stand out as some of the most advanced technologies in the world.
 - Emerging technologies can enhance human productivity but also create unforeseen risks.
 - As we develop new tech, cybersecurity practices must also evolve.
 - In the long run, AGI has the potential to be the most significant technological innovation.



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BIG DATA TECHNOLOGIES AND TOOLS

- **Programming Models**
 - Batch vs. Incremental vs. Real-Time
 - Structured vs. unstructured data
 - Declarative vs. generalist programming
- **Data Storage**
 - Distributed file systems (e.g. HDFS)
 - Relational database.
 - NoSQL databases (e.g. key-value stores, document archives)
 - Integration of multiple data IGNOREEs (e.g. Hive)



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BIG DATA TECHNOLOGIES AND TOOLS

- **Generic processing platforms**
 - Infrastructure: context, properties and implications
 - Map-reduce model and support platform (e.g. Hadoop)
 - Second-generation platforms (e.g. Pig, Spark)
- **Processing in specific domains**
 - Libraries for machine learning (e.g. Spark MLlib)
 - Graph processing platforms (e.g. GraphX)
- **Introduction to Real-Time Processing Platforms**
 - Data sources (e.g. Flume, Kafka)
 - Data models: micro-batches vs. continuous flows
 - Processing platforms (e.g. Storm, Spark Streaming)

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BIG DATA TECHNOLOGIES AND TOOLS

- **Processing engines** - Spark, Hadoop MapReduce and stream processing platforms such as Flink, Kafka, Samza, Storm and Spark's Structured Streaming module.
- **Storage repositories** - Hadoop Distributed File System and cloud object storage services such as Amazon Simple Storage Service and Google Cloud Storage.
- **NoSQL databases** - Cassandra, Couchbase, CouchDB, HBase, MarkLogic Data Hub, MongoDB, Redis and Neo4j.
- **SQL query engines** - Drill, Hive, Presto and Trino.
- **Data lake and data warehouse platforms** - Amazon Redshift, Delta Lake, Google BigQuery, Kylin and Snowflake.
- **Commercial platforms and managed services** - Amazon EMR, Azure HDInsight, Cloudera Data Platform and Google Cloud Dataproc.